

ENGINEERING ECONOMICS

Annual Cash Flow Analysis & Loan Amortization

Evaluating Alternatives with Equivalent Uniform Amounts

Presentation Overview



Part A

Foundation & Criteria
等额年金基础与经济
准则



Part B

EUAC Formulas & Derivation
年金成本公式与推导



Part C

Loan Amortization & Excel
贷款摊销与Excel应用



Part D

Applications & Summary
工程应用场景与总结

Part A

Foundation & Economic Criteria

Understanding the core concepts of annual cash flows

What is Annual Cash Flow Analysis?

等额年金分析法 (Annual Cash Flow Analysis) 是一种工程经济学核心技术，它将项目的所有现金流转化为一组等额的期末年金 (Equivalent uniform end-of-period amounts)。



The Ultimate Advantage:

(相比PW分析的最大优势)

当比较不同寿命期 (Different useful lives) 的备选方案时，年金分析法不需要寻找最小公倍数 (LCM) 或设定共同分析期。因为等额年金已经代表了一个周期的“年均成本”，可以直接在不同寿命方案间进行公平比较。

Key Metrics of Annual Cash Flow

The results of this analysis are expressed as three mathematically related metrics:



EUAC

Equivalent Uniform Annual Cost
(等额年金成本)

将所有初始投资、维护费和残值平摊到每年的净成本。分析目标通常是 **Minimize (最小化)**。



EUAB

Equivalent Uniform Annual
Benefit (等额年金收益)

将项目产生的未来所有收益平摊到每年的净收入。分析目标通常是 **Maximize (最大化)**。



EUAW

Equivalent Uniform Annual Worth
(等额年金净值)

$$\text{EUAW} = \text{EUAB} - \text{EUAC}$$

代表项目的年均净利润。分析目标始终是 **Maximize**。

Economic Criteria (Table 6-1)

根据投入 (Input) 和产出 (Output) 是否固定，选择正确的决策准则：

Input / Output Condition (情况)	Situation Description (场景描述)	Economic Criterion (经济准则)
Case 1: Neither Fixed	最常见的普遍情况。投入资金不固定，预期收益也不固定。	↑ Maximize EUAW (EUAB - EUAC)
Case 2: Fixed Input	项目的可用预算或资源总量已被严格限制 (Budget is fixed)	↑ Maximize EUAB (获取最大化年收益)
Case 3: Fixed Output	必须要完成一个既定任务或实现固定收益 (Task is fixed)	↓ Minimize EUAC (以最低年成本完成)

Example 6-5

Given: 一家公司正在考虑安装两台设备中的哪一台以降低成本。两台设备的使用寿命均为5年，无残值。设备A的成本为1000美元，预计每年可节省300美元。设备B的成本为1350美元，第一年可节省300美元；此后，节省金额每年增加50美元，即第二年节省350美元，第三年节省400美元，依此类推。在利率为7%的情况下，公司应购买哪台设备？

DeviceA:

$$\begin{aligned} EUAW &= -1000 (A/P, 7\%, 5) + 300 \\ &= -1000 (0.2439) + 300 \\ &= \$56.1 \end{aligned}$$

DeviceB:

$$\begin{aligned} EUAW &= -1350 (A/P, 7\%, 5) + 300 + 50 (A/G, 7\%, 5) \\ &= -1350 (0.2439) + 300 + 50 (1.865) \\ &= \$64.00 \end{aligned}$$

Part B

EUAC Formulas & Derivation

Capital recovery and equivalent uniform annual cost

Definition of EUAC (Capital Recovery Cost)

EUAC 将拥有资产的总成本（包括初始投资 P 和期末的残值 S ）在考虑资金时间价值的前提下，平摊为每年的等额成本。

- ✓ 也被称为资金回收成本 (Capital Recovery Cost)。
- ✓ 它代表了为了证明该投资的合理性，资产每年必须产生的最低年收入。
- ✓ 管理者通常习惯以“年度”为单位思考预算，因此 EUAC 比总现值(PW)更具实用指导意义。



The Transformation

Lump Sum Investment (P) at Year 0
+ Lump Sum Salvage (S) at Year n



Smooth Annual Payments (A)

The Three Equivalent EUAC Formulas

当存在初始支出 P 和期末残值 S 时，年度成本有三种等效计算方式（结果完全一致）：

Formula 1

MOST COMMON

$$EUAC = P(A/P, i, n) - S(A/F, i, n)$$

将初始成本年度化，减去残值的年度化收益。

Formula 2

$$EUAC = (P - S) (A/F, i, n) + Pi$$

将未回收成本加上初始投资产生的利息。

Formula 3

$$EUAC = (P - S) (A/P, i, n) + Si$$

将折旧金额年度化，加上残值产生的利息。

Derivation of Formula 1 / 公式一的推导过程

让我们逐步推导最常用的公式1，理解其背后的资金时间价值逻辑：

Step 1: Convert Initial Cost (P) to Annual Amount

将时间0点的初始成本 P 转化为等额年金，使用资金回收系数：

$$A_P = P * (A/P, i, n)$$

Step 2: Convert Salvage Value (S) to Annual Benefit

将时间 n 点的期末残值 S 转化为等额年金，使用偿债基金系数（由于它是收入，将抵扣成本）：

$$A_S = S * (A/F, i, n)$$

Step 3: Calculate Net EUAC

$$EUAC = A_P - A_S = P * (A/P, i, n) - S * (A/F, i, n)$$

EUAC Calculation Example-1

Given: Initial Cost $P = \$1,000$, Salvage $S = \$200$, $n = 10$ years, $i = 7\%$.



$$\begin{aligned} EUAC &= 1000 * (A/P, 7\%, 10) - 200 * (A/F, 7\%, 10) \\ &= 1000 * (0.1424) - 200 * (0.0724) \\ &= 142.40 - 14.48 \\ &= 127.92 \end{aligned}$$

Comparing Alternatives (Equal vs. Different Lives)

Equal Lives (相同寿命)

When alternatives have the same life, compare EUAC directly.

Both machines have a 5-year life:

Machine A: EUAC = \$1,200

Machine B: EUAC = \$1,100

✓ Select Machine B (Lower Cost)

Different Lives (不同寿命)

Annual cash flow analysis can directly compare alternatives with different lives.



Key Assumption:

Identical Replacement - 假设每个方案在其寿命结束时，都会被一个完全相同的新单元替换。

Because of this assumption, the EUAC represents the perpetual annual cost. No LCM timeline is needed!

Example 2: Comparing Alternatives (Different Lives)

Problem: At 12% interest, which machine should be selected using EUAC analysis?

Machine A (Life = 4 years)

-  Cost (P): \$2,000
-  Annual O&M: \$1,200
-  Salvage (S): \$800

$$\begin{aligned} \text{EUAC} &= 2000A / P, 12\%, 4 + 1200 \\ &\quad - 800A / F, 12\%, 4 \\ &= 20000.3292 + 1200 - 8000.2092 \\ &= \$1,691.04 \end{aligned}$$

Machine B (Life = 6 years)

-  Cost (P): \$7,000
-  Annual O&M: \$900
-  Salvage (S): \$1,200

$$\begin{aligned} \text{EUAC} &= 7000A / P, 12\%, 6 + 900 \\ &\quad - 1200A / F, 12\%, 6 \\ &= 70000.2432 + 900 - 12000.1232 \\ &= \$2,454.56 \end{aligned}$$

 Decision: Select Machine A (Minimize EUAC).

Example 3: Investment Decision (EUAW)

A company is considering purchasing a machine. Should the machine be purchased? (Use EUAW analysis).



Cost (P): \$50,000



Annual Savings: \$12,000



Life (n): 6 years



Salvage (S): \$5,000



Interest (i): 15%

EUAC

$$= 50000(A/P, 15\%, 6) - 5000(A/F, 15\%, 6)$$

$$= 500000.2642 - 50000.1142 = \$12,639$$

EUAW

$$= \text{Savings} - \text{EUAC}$$

$$= 12000 - 12639 = -\$639$$



Decision: EUAW < 0. Do NOT purchase.



Part C

Loan Amortization & Excel

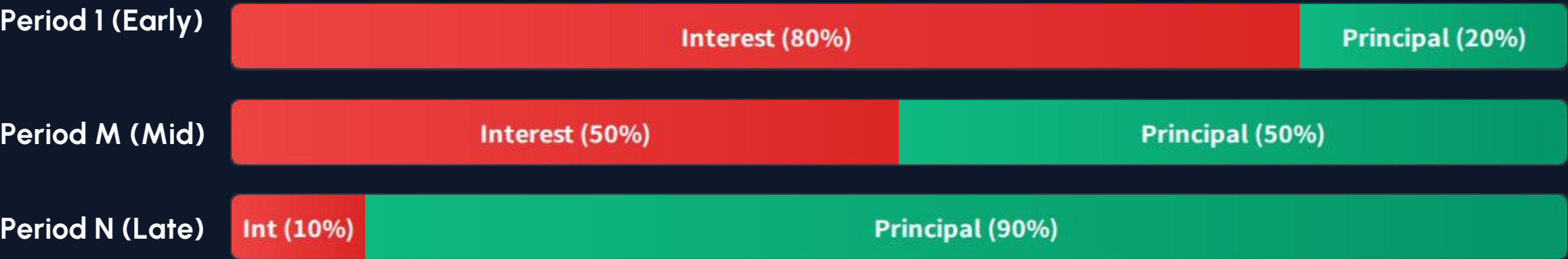
Dissecting debt structure and financial modeling

What is Loan Amortization? / 什么是贷款摊销？

Loan amortization is the process of paying off a loan through regular payments that cover both principal and interest.

Each Payment = Interest (利息) + Principal (本金)

 Key Insight (核心洞察):
在还款初期，大部分资金用于支付利息；随着本金余额的减少，后期还款中偿还本金的比例会逐渐增大。



Example 4: Amortization Schedule (Example 6-11)

Problem: A graduating student spends \$2,400 on new furniture using 6-month financing at 6% per year nominal (0.5% per month). Determine the payment and amortization table.

$$A = 2400A / P , 0.5 \% , 6 = 24000.16959 = \$ 407.03$$

Month	Interest (0.5%)	Total Payment	Principal Paid	Ending Balance
0	-	-	-	\$2,400.00
1	\$12.00	\$407.03	\$395.03	\$2,004.97
2	\$10.02	\$407.03	\$397.01	\$1,607.96
3	\$8.04	\$407.03	\$398.99	\$1,208.97
4	\$6.04	\$407.03	\$400.99	\$807.98
5	\$4.04	\$407.03	\$402.99	\$405.00
6	\$2.03	\$407.03	\$405.00	\$0.00

Amortization Schedule / 摊销计划表

A structured table demonstrating how a \$10,000 loan at 12% nominal rate (1% monthly) is paid off over exactly 3 months with equal payments of \$3,400.22.

Period (n)	Beginning Balance	Interest (1%)	Principal Paid	Total Payment	Ending Balance
1	\$10,000.00	\$100.00	\$3,300.22	\$3,400.22	\$6,699.78
2	\$6,699.78	\$67.00	\$3,333.22	\$3,400.22	\$3,366.56
3	\$3,366.56	\$33.67	\$3,366.56	\$3,400.23*	\$0.00

** Slight rounding adjustment applied to final payment to zero out balance.*

Finding Balance After k Payments / 计算剩余余额

在任意第 k 期后，计算未偿还的贷款余额 B_k 有两种完全等效的方法：

Method 1: Retrospective (回溯法)

Outstanding balance = Total loan rolled over to present - Total paid back rolled over to present.

$$B_k = P(i + i)^k - A\left[\frac{(i + i)^k - 1}{i}\right]$$

假设过去 k 个月一分钱没还的本息和，减去过去 k 个月你已经还掉的钱的终值。

Method 2: Prospective (前瞻法)

The balance due equals the present worth of the **remaining payments**.

$$B_k = AP / A, i, N - k$$

直接把未来还剩下的 $(N-k)$ 期等额还款，折现到当前的 k 时刻。

Financial Excel Functions / 财务Excel函数

在实际工程与金融建模中，我们严重依赖电子表格软件（如 Excel/Google Sheets）来处理复杂的摊销计算。

```
// 1. Calculate periodic payment (已知P求A)
```

```
=PMT (rate, nper, pv)
```

```
// 2. Calculate remaining balance after k payments (计算Bk)
```

```
=FV (rate, k, pmt, -pv)
```

```
/* Note: Use negative pv to represent initial cash received */
```

```
// 3. Calculate cumulative interest paid between periods (累计支付利息)
```

```
=CUMIPMT (rate, nper, pv, start_period, end_period, 0)
```

```
// 4. Calculate cumulative principal repaid between periods (累计偿还本金)
```

```
=CUMPRINC (rate, nper, pv, start_period, end_period, 0)
```

Car Loan Analysis Example

A car loan of \$25,000 is taken at 9% annual interest for 5 years with monthly payments. Calculate:

- a) The monthly payment
- b) The total interest paid over the life of the loan
- c) The balance remaining after 2 years

Method-1:

a) Monthly Payment (A):

$$A = 25000 * (A/P, 0.75\%, 60) = 25000 * [0.0075(1.0075)^{60} / (1.0075^{60} - 1)] \rightarrow \$518.96$$

or $A = \text{PMT}(0.09/12, 60, 25000)$

b) Total Interest Paid:

$$= (518.96 * 60) - 25000 \rightarrow \$6,137.60$$

c) Balance Remaining After 2 Years (k=24):

$$= \text{FV}(0.0075, 24, -518.96, 25000) \rightarrow \$16,319.60$$

Method-2:

Excel

Example 5: Finding Remaining Balance (Example 6-12)

Problem: A car is purchased with a 48-month, 9% nominal loan (0.75% monthly) with an initial balance of \$15,000. What is the balance halfway through (after 24 months)?

Step 1: Calculate the standard monthly payment (A)

$$A = 15000A / P, 0.75\%, 48 = 150000.024885 = \$373.28$$

Step 2: Apply Prospective Method for B_{24}

The balance due equals the present worth of the **remaining 24 payments**.

$$B_{24} = 373.28P / A, 0.75\%, 24 = 373.2821.889 = \$8,170.78$$

Notice that halfway through the time period, you still owe 54.5% of the original \$15,000 principal due to front-loaded interest.